

☒ EASY

☐ ALGEBRA

Algebra

10 questions · ~15 min

12

Q1

ALGEBRA

Leo goes to a packing store to buy containers and tape. Leo has \$ 15. Each container costs \$ 1.87 and each roll of tape costs \$ 2.40. Which inequality represents the relationship between the number of containers, c , and the number of rolls of tape, t , Leo can buy?

- A. $1.87c + 2.40t \leq 15$
- B. $1.87c + 2.40t \geq 15$
- C. $2.40c + 1.87t \leq 15$
- D. $2.40c + 1.87t \geq 15$

Tony spends \$80 per month on public transportation. A 10-ride pass costs \$12.50, and a single-ride pass costs \$1.50. If g represents the number of 10-ride passes Tony buys in a month and t represents the number of single-ride passes Tony buys in a month, which of the following equations best represents the relationship between g and t ?

- A. $g + t = 80$
- B. $g + t = 1.50 + 12.50$
- C. $1.50g + 12.50t = 80$
- D. $12.50g + 1.50t = 80$

$$8x = 88$$

What value of x is the solution to the given equation?

- A. 11
- B. 80
- C. 96
- D. 704

For a training program, Juan rides his bike at an average rate of 5.7 minutes per mile. Which function m models the number of minutes it will take Juan to ride x miles at this rate?

A. $mx = \frac{x}{5.7}$

B. $mx = x + 5.7$

C. $mx = x - 5.7$

D. $mx = 5.7x$

Q5

GRID-IN ALGEBRA

$$\begin{aligned}3x + y &= 29 \\ x &= 2\end{aligned}$$

If (x, y) is the solution to the given system of equations, what is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$2l + 2w \leq 27$$

A rectangle has length l and width w . The inequality gives the possible values of l and w for which the perimeter of this rectangle is less than or equal to 27. Which statement is the best interpretation of $l, w = 8, 3$ in this context?

- A. If the rectangle has length 3 and width 8, its perimeter is less than or equal to 27.
- B. If the rectangle has length 8 and width 3, its perimeter is less than or equal to 27.
- C. If the rectangle has length 3 and width 8, its perimeter is greater than or equal to 27.
- D. If the rectangle has length 8 and width 3, its perimeter is greater than or equal to 27.

A producer is creating a video with a length of 70 minutes. The video will consist of segments that are 1 minute long and segments that are 3 minutes long. Which equation represents this situation, where x represents the number of 1-minute segments and y represents the number of 3-minute segments?

- A. $4xy = 70$
- B. $4x + y = 70$
- C. $3x + y = 70$
- D. $x + 3y = 70$

Q8

GRID-IN

ALGEBRA

If $3x + 2 = 8$, what is the value of $9x + 6$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Hana deposited a fixed amount into her bank account each month. The function $f(t) = 100 + 25t$ gives the amount, in dollars, in Hana's bank account after t monthly deposits. What is the best interpretation of 25 in this context?

- A. With each monthly deposit, the amount in Hana's bank account increased by \$ 25 .
- B. Before Hana made any monthly deposits, the amount in her bank account was \$ 25.
- C. After 1 monthly deposit, the amount in Hana's bank account was \$ 25.
- D. Hana made a total of 25 monthly deposits.

$$x = 4$$

$$y = 5 - x$$

The solution to the given system of equations is x, y . What is the value of y ?

A. 1

B. 4

C. 5

D. 9